[CH 10] The Shaping of the Personal Computer

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- The usual portrayal of a few companies and individuals as visionaries and of the established computer industry as deservedly extinct is misleading

Radio Days

- The PC was shaped by the interaction of various cultural forces and commercial interests
- This can be compared to the development of radio

- Initially, in the 1890s, radio was just a scientific novelty. There were no actual applications for it yet
- The first application was wireless telegraphy
- When Guglielmo Marconi transmited the letter S repeatedly across the atlantic, this proved that it was possible and telegraph y firms flocked to him
- As wireless telegraphy started being used by companies and the government, private hobbyists started showing up
- They were the first amateur broadcaster, operators and listeners
- Without them, the radio broadcasting industry might never have developed
- Once the first few were established, there were more listeners, which justified more stations, which attracted more listeners , etc
- This created a demand for radio hardware and the radio-set industry was born
- RCA sold $80 million worth of radio sets in 4 years starting with 1921
- Existing firms like Westinghouse and General Electric began making radio sets, competing fiercely with start-ups like Amrad, De Forest, Stromberg-Carlson, Zenit, etc

- There are 3 important points:
  - Radio came from a new enabling technology, whose long term importance had not yet been recognized
  - A crucial set of actors in the transformation were amateurs
  - Once radio broadcasting was established, it was quickly dominated by a few giant firms
- In computers:
  - The enabling technology is the microprocessor
  - The computer amateur played an important but underappreciated role
  - PCs spawned a major industry

Microprocessors

- The enabling technology for the PC
- Developed 1969-1971 in the semiconductor firm Intel (it was, like many other developments, invented independently in more than one place, but Intel was the most important)
- Intel was founded 1968 by Robert Noice and Gordon Moore, vice presidents of Fairchild Semiconductor

- 1969 Intel was approached by japanese calculator manufacturer Busicom to develop a chip set for a new scientific calculator
- Ted Hoff and his coworkers were given the job of designing it
- Hoff set off to design a general-purpose chip that could be programmed with the calculator functions, instead of specially designed logic chips
- This chip would, in fact, be a rudimentary computer
- The 4004 chip was delivered to Busicom in 1971, but the company fell victim to the calculator price wars of the early 1970s
- Intel got the rights to market the chip on its own account
- It announced it as a "microprogrammable computer on a chip" in November 1971. It sold for $1,000 at first
- The 4004 was replaced by the 8008, that could process 8 bits at a time
- The 8080 came out in 1974 and became the basis for several personal-computer designs
- Motorolla was producing the Motorolla 6800, Zilog the Z80 and MOS Technology the 6502
- Due to competition, the price dropped to $100
- The first real personal computer was the Apple II

Computer Hobbyists and "Computer Liberation"

- Most hobbyists had some electronics experience, be it from working with or around computers, or from radio
- Their enthusiasm was usually sparked by an interaction with minicomputers

    - These were in the range of 20,000 USD, way too expensive
- Some of the hardware was readily available in government surplus shops (ex: teletypes), but the processor remained hard to get and expensive

The Altair 8800

- The first microprocessor based computer
- Announced in 1975 in the magazine *Popular Electronics*
- The first with a price so low that it could be realistically bought by an individual
- Cost $397
- A minicomputer in all other respects
- It had no display, no keyboard and only 256 bytes of memory
- It could only programmed by entering programs in pure binary through switches on the front of the machine
- It was produced by New Mexico electronics kit supplier MITS (Micro Instrumentation Telemetry Systems)
- Small-time entrepreneurs developed add-on boards to add more memory, teletyles, audiocassette recorders for data storage, etc

- When the Altair 8800 came out, Bill Gates and Paul Allen approached Ed Roberts at MITS to develop a BASIC programming system for the machine
- Gates and Allen formed Micro-Soft and delivered a BASIC programming system in 6 weeks (February 1975)
- Allen became software director at MITS

- Hundreds of small firms soon entered the microcomputer software business, and Microsoft was not the most prominent
- In Q1 of 1975, MITS received over $1 million orders for Altair 8800s
- At its first worldwide conference, Gates spoke against software piracy. He was met with hostility but his position was eventu ally accepted.
    - This transformed the PC into an economic artifact
- In 2 years (1975-1977) the computer became a consumer product

The Rise of Apple Computer

- Founded by Stephen Wozniak and Steve Jobs
- Wozniak had no degree or formal qualifications, but he was so talented that he was hired in the calculator division of Hewlet t-Packard
- After learning of microprocessors and the Altair 8800, Wozniak and Jobs built a computer based on the MOS Technology 6502 chip and called it the "Apple"
- Jobs recognized that the microcomputer could become a product for a broad market if packaged appropriately
- It needed a display, a keyboard, permanent storage and had to be self-contained and pluggable into a regular outlet like any other appliance. This was the Apple II

- At the West Coast Computer Faire in April 1977, the Apple II and the Commodore Business Machines Commodore PET were launched
- The PET was cheaper but had a narrow specification and no add-on options
- The Apple II was more expensive ($1,298) but was more fully featured
- August 1977, Tandy announced the TRS-80 computer for $399
- The apple II appealed to hobbyists, but Jobs wanted the machine to be an appliance in the home, a commonplace item
- The group of users that would use the PC would be defined by the available software

- 1976 there were a handful of PC software firms making "system" software
- The most popular were Microsoft's BASIC programming language and Digital Research's CP/M operating system
- 1977 it was still a small business. Microsoft had 5 employees and $500,000 annual sales
- With the appearance of the Apple II, the PET and the Tandy TRS-80, the market for software applications took off
- There were 3 main sections:
    - Games
        - Initially the biggest market
    - Education
        - Second biggest market
        - Schools and colleges were the first to buy large volumes of PCs
        - Software was needed to learn mathematics, simulation programs needed for teaching, programs for business

        - games, language learning and music
        - Early on, this development was done by students and teachers and the final product was quite poor quality
        - Some were developed through research grants, which meant they were given away for free or sold on a nonprofit basis
        - The market developed haphazardly
    - Business
        - The first widely used acceptance was VisiCalc spreadsheet, a financial analysis tool developed by Harvard MBA student Daniel Bricklin
        - VisiCalc user 25,000 bytes of memory, which was a LOT
        - The PC had the advantage of responding VERY fast, almost instantaneously
        - Users of VisiCalc experienced for the first time the freedom of having their own computer
- Word processing on PCs only developed in the 1980s
    - First generation PCs displayed 40 uppercase characters in a line
    - Good quality printers were expensive
    - By 1980, new computers that could display 80 uppercase and lowercase characters were on the market
    - This, along with the availablilty of low-cost, reasonable qualoty japanese printers greatly helped the word processing market
    - Micro-Pro was the first firm to produce successful word-processing software, founded 1978 by Seymour Rubinstein
        - 1978 they produced WordMaster
        - 1979 they produced WordStar
        - They sold nearly 1 million copies of their software and were a $100 million / year business

The IBM PC and the PC Platform

- Once the PC was clearly defined as a business machine in the 1980s, IBM reacted with surprising speed
- The proposal came from Willian C. Lowe, who headed the "entry level systems" division
- Lowe recommended following industry trends:
    - Oursourcing components it did not already have in production
    - Using the regular retail channels instead of its specialised direct sales force
- His advice was agreed to by top management
- Within 2 weeks he was authorized to build a prototype. It had to be ready within 12 months. This was Project Chess

- Because of their late entry, they had a few advantages:
    - 16-bit microprocessors (2nd generation) was the most important - They chose the Intel 8088
- IBM was the world's largest software developer, but they did not have the skills for PC software.
- They initially approached Gary Kildall of Digital Research, but he blew the opportunity
- They then turned to Bill Gates and Paul Allen at Microsoft and signed the deal
- Gates obtained a suitable piece of software from Seattle Computer Products for $30,000 and developed it into MS-DOS
- This was bundled with every IBM PC and got Microsoft $10-$50 for every unit sold

- Fall of 1980 the PC was finished and named the "Acorn" internally
- Don Estridge was put in charge of the project and he arranged factory deals for the components that IBM did not make themselves
- They arranged contracts with the Sears Company to sell their machine, along with ComputerLand
- Upon launch on 12 August 1981, the IBM Personal Computer (as it was named) became a runaway success
- The IBM logo and presence had legitimized the Personal Computer
- The fully equipped machine cost $2,880 and demand was so high that within a few days IBM decided to quadruple production

- The IBM PC became an industry standard in 1982-1983
- Clone machines started being manufactured
    - Among the most successful clone manufacturers was Compaq. In its first full year of business, it achieved sales of $110 million
- Thousands of software programs were published for the IBM PC
- Time magazine nominated the PC as Man of the Year

Dell Computer and Process Innovations

- Michael Dell, a University of Texas student, realised that IBM was selling a computer made out of $700 components for almost $3,000
- He started upgrading stripped-down PC's that he bought at a discount and was soon selling $50,000 - $80,000 worth of

computers a month

- He incorporated as PC's Limited
- 1985 Dell hired Jay Bell to create a PC based on Intel's 80286 processor - it was called the 286
- Dell sold by mail order to cut out the retail middleman
- PC's Limited often offered cheaper and better-value machines
- Dell targeted the corporate market from the start
- He allowed customers to choose their own specifications for disk drives, memory and other features, known as mass customization
- 1988 the company became Dell Computer Corporation
- 1986 they had gained publicity for having built the world's fastest PC

- Nearly all PC's built by any major manufacturer were using a Microsoft OS and Microsoft applications software
- 80% of all PC's were using Intel microprocessors
- Almost all companies who did not switch to the IBM standard went out of business. The notable exception is Apple Computer, who decided to compete not with chearper hardware, but with better software